

Qingyang Zhou

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<https://esperarr.github.io/CV/>

EDUCATION

Tsinghua University September 2022 - Present
Xinya College, Tsinghua University: <https://www.xyc.tsinghua.edu.cn/index.htm>
Department of Computer Science and Technology, Tsinghua University: <https://www.cs.tsinghua.edu.cn/>
GPA 3.6/4.0
Selected A+ professional compulsory course: Introduction to Complex Analysis, Media Computing, Database Special Topic Training;

RESEARCH PUBLICATION

"Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation"

Yuwei Zhou, Xin Wang, Hong Chen, Yipeng Zhang, **Qingyang Zhou**, Wenwu Zhu
08 Aug 2024 (modified: 10 Dec 2024) Submitted to AAAI 2025
<https://openreview.net/pdf?id=kiGUqX6Gct>

"Towards Mutually Illuminative Collaborative Human-AI Deliberate Discussion"

Zipeng Zhang, Shiwei Wu, Ran Ran, **Qingyang Zhou**, Ranrui Ma, Jiaye Leng, Jian Zeng, Zhenhui, Chun Yu
09 April 2025 Submitted to UIST 2025
<https://new.precisionconference.com/uist25a/coauthor/subs/1831>

INTERNSHIP/TRAININGS

Client Development Engineer Intern at Tencent (Project MAX) May 2026 - August 2026
Tencent, Project MAX Team

- Optimized NavMesh pathfinding to improve agent movement smoothness, resolving instability on terrain bumps and convex corner edges.
- Implemented NPC auto-navigation and pathfinding behavior using custom FixNavMesh algorithms.
- Developed a GM panel for debugging and performance stats toggling in the battle UI, and built the game control panel UI for the 2V2 battle mode.

scientific research internship at Multimedia and Network Big Data Lab July 2023 - July 2024

Tsinghua University Multimedia and Network Big Data Lab: <https://mn.cs.tsinghua.edu.cn/index/index.html>

- Conducted study and research on diffusion models and personalized image generation, within the project "Video Generation System Combining Large Language Models with Diffusion Models".
- Assisted in publishing the paper: "Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation".

scientific research internship at Pervasive Human Computer Interaction Lab January 2025 - April 2025

Tsinghua University Pervasive Human Computer Interaction Laboratory: <https://pi.cs.tsinghua.edu.cn/>

- Conducted research on aligning human and AI thought processes.
- Developed a human-AI interactive writing assistant to support collaborative idea generation.
- Contributed to the research and UIST 2025 submission of the paper "Towards Mutually Illuminative Collaborative Human-AI Deliberate Discussion".

scientific research internship at Information Retrieval LabOctober 2024 - Present

Information Retrieval Lab at Tsinghua University: <http://www.thuir.cn/>

- Focused on improving traditional recommendation models (such as FM, DeepFM, AFM).
- Incorporated SAE plugins into these models to enhance their interpretability, controllability, and transparency.

POSITION OF RESPONSIBILITY

Secretary of the Intelligent Agent Department of the Student Science and Technology AssociationFebruary 2023 - Present

Tsinghua University CST Department, Student Science and Technology Association: <https://net9.org/home/>

- Participated in the game development and event organization for the Intelligent Agent Competition.
- Mainly responsible for the development of the Intelligent Agent-Based Game 'Generals' Project in the 28th Tsinghua University Intelligent Agent Competition.

Volunteer Tutor at Tsinghua Q&A WorkshopSeptember 2025 - Present

Tsinghua University, Center for Student Learning and Development: <https://learning.tsinghua.edu.cn/zxgk/ywjs.htm>

- Provided tutoring for programming and computer science courses, including C/C++ programming, data structures, and algorithms.
- Assisted students with debugging, algorithm design, and understanding core programming concepts during regular tutoring sessions.

Tutor of Tsinghua University Science and Engineering Summer Camp July 2024 - August 2024

Tsinghua University 2024 Science and Engineering Summer Camp I6 Class Tutor

- Responsible for event organization, answering professional questions, and other related tasks.
- Led and organized engineering-related activity projects, such as intelligent robot design and card-bridge construction.

PROJECTS

Path Tracing RendererJune 2026

Personal Graphics Project: <https://github.com/Esperarr/TinyPathTracer>

- A physically-based path tracer supporting both Whitted-style ray tracing and Monte Carlo path tracing with Russian Roulette for unbiased termination.
- Implements PBR materials with GGX microfacet importance sampling (isotropic & anisotropic), Cook-Torrance BRDF, and Fresnel Schlick; covers diffuse, mirror, glass, and glossy surfaces.
- Features BVH acceleration for 100k+ face meshes, NEE + MIS with power heuristic weighting, depth of field, MSAA anti-aliasing, gamma correction, and OpenMP parallelization.

Intelligent Agent-Based Game 'Generals'September 2023 - May 2024

Major Project in the 28th Tsinghua University Intelligent Agent Competition: <https://www.saiblo.net/game/35>

- A Turn-Based Strategy Game Project developed for the competition.
- Responsible for writing backend logic, unit tests, debugging, creating the SDK, authoring documentation, and guiding participants through the competition.

SKILLS

Languages	Chinese (Native), English (Fluent), Spanish (Basic)
Programming	C/C++, Python, JavaScript/TypeScript, Java, Rust, Verilog/SystemVerilog
Graphics / Tools	OpenGL, GLSL, Unity, PyTorch, Git, Linux, Docker, LaTeX

EXTRA-CURRICULAR

- Served as a volunteer for Tsinghua University's Tianjin recruitment teamAugust 2023
- Served as the Publicity Department Executive in Xinya College. September 2022 - September 2023